

RESEARCH ARTICLE

Structure–function coupling reveals the brain hierarchical structure dysfunction in Alzheimer's disease: A multicenter study

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Funding information

the Science and Technology Innovation 2030 Major, Grant/Award Number: 2022ZD0211600; the Beijing Municipal Natural Science Foundation, Grant/Award Number: 7244519; Fundamental Research Funds for the Central Universities, Grant/Award Number: 2021XD-A03; National Natural Science Foundation of China, Grant/Award Numbers: 62333002, 82172018; Beijing Nova Program, Grant/Award Number: 20220484177; Science and Technology Project of Tianjin Municipal Health Committee, Grant/Award Number:

Abstract

BACKGROUND: Alzheimer's disease (AD) is a neurodegenerative condition characterized by cognitive decline. To date, the specific dysfunction in the brain's hierarchical structure in AD remains unclear.**METHODS:** We introduced the structural decoupling index (SDI), based on a multi-site data set comprising functional and diffusion-weighted magnetic resonance imaging data from 793 subjects, to assess their brain hierarchy.**RESULTS:** Compared to normal controls (NCs), individuals with AD exhibited increased SDI within the posterior superior temporal sulcus, insular gyrus, precuneus, hippocampus, amygdala, postcentral gyrus, and cingulate gyrus; meanwhile, the patients with AD demonstrated decreased SDI in the frontal lobe. The SDI in those regions also showed a significant correlation with cognitive ability. Moreover, the SDI was a robust AD neuroimaging biomarker capable of accurately distinguishing diagnostic status (area under the curve [AUC] = 0.86).

Yibao Sun, Pan Wang, and Kun Zhao contributed equally to this study.

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TJWJ2022MS032; Tianjin Key Medical
Discipline (Specialty) Construction,
Grant/Award Number: TJYXZDXK-052B

DISCUSSION: Our findings revealed the dysfunction of the brain's hierarchical structure in AD. Furthermore, the SDI could serve as a promising neuroimaging biomarker for AD.

KEYWORDS

Alzheimer's disease, hierarchical structure, multi-modal brain network, multi-site, structural decoupling index

Highlights

- This study utilized multi-center, multi-modal data from East Asian populations.
- We found an increased spatial gradient of the structure decoupling index (SDI) from sensory-motor to higher-order cognitive regions.
- Changes in SDI are associated with energy metabolism and mitochondria.
- SDI can identify Alzheimer's disease (AD) and further uncover the disease mechanisms of AD.

1 | INTRODUCTION

Alzheimer's disease (AD) is a neurodegenerative disorder characterized by various impairments including memory, emotional, cognitive, and behavioral deficits.¹⁻³ The human brain functions as a complex network and plays a crucial role in constraining brain function.⁴⁻⁶ The hierarchical structure of the brain network serves as the basic form of organization for translating and integrating information.^{7,8} Dysfunction of the brain's hierarchical organization leads to abnormal structural damage and altered functional integration among different brain regions through disconnection mechanisms.^{6,9} Understanding these changes in hierarchical structure and functional connectivity is essential for investigating the underlying mechanisms of AD. Better understanding these structure and function alterations will shed light on the potential targets for therapeutic interventions to mitigate the effects of the AD on brain function and cognition.

Quantifying the brain's hierarchical structure is a challenge in neuroscience.¹⁰ Previous studies have demonstrated that the role of hierarchical organization lies in elucidating how brain structure influences and constrains brain function.¹¹ Thus the decoupling of function and structure offers promising insight for exploring the intricate hierarchical structures of the brain. Several studies have employed a straightforward regional spatial correlation to investigate the relationship between brain structure and function.¹²⁻¹⁵ However, the coupling of function and structure exhibits a complex phenotype, which may not comprehensively characterize the complexity of the function-structure relationship by using only linear statistical methods.

To address these challenges, Preti & De Ville have proposed an innovative measure known as the structural decoupling index (SDI) to quantify the degree of regional structure-function dependency.¹¹ SDI combines network science¹⁶ and graph signal processing¹⁷ to link brain activity signals with the underlying white matter topology. Furthermore, SDI is a novel neuroimaging biomarker for the assess-

ment of the relationship between structural and functional activity. Consequently, the SDI holds promising potential for quantifying the brain's hierarchical structure, thereby enhancing our comprehension of brain communication mechanisms that underpin the diverse behavioral domains and uncovering the potential genetics.^{11,18} Therefore, the dysfunction of the brain's hierarchical structure in AD can be quantified through the SDI.¹¹

The primary aim of this study is to explore the alteration of the brain's hierarchical structure in AD. For this purpose, we first computed the individual SDIs by combining the resting-state functional magnetic resonance imaging (rs-fMRI) and diffusion tensor imaging (DTI) in a large multi-site data set encompassing a cohort of 284 patients with AD, 254 individuals with mild cognitive impairment (MCI), and 255 cognitively normal controls (NCs) from seven sites. Then we performed a mega-analysis framework to investigate the robust alteration of the SDI in AD. In addition, we elucidated the underlying mechanism of SDI alterations in AD via gene enrichment analysis. Finally, to evaluate the potential of SDIs as a neuroimaging biomarker for AD, we employed a support vector machine (SVM) model for AD classification and clinical score prediction (Figure 1).

2 | MATERIALS AND METHODS

2.1 | Participants and image acquisition

This study included 793 participants (284 individuals with AD, 254 with MCI, and 255 NCs) with paired fMRI scans, DTI scans, and demographic and psychological information from seven sites of the Multi-Center Alzheimer's Disease Imaging (MCADI), (Table S1). The MRI scanner and acquisition protocol information of fMRI and DTI data are listed in Tables S2 and S3. Detailed information can be found in our previous studies.¹⁹⁻²² This study was consistent with the Principles of the

Declaration of Helsinki and approved by the medical research ethics committee and institutional review board (detailed information can be found in [Method S1](#)).

2.2 | Data preprocessing

All the rs-fMRI scans were pre-processed using the Brainnetome fMRI Toolkit (<http://brant.brainnetome.org>), which included the following steps: (1) slice-timing correction; (2) realignment to the first volume; (3) normalization (Montreal Neurological Institute [MNI] space with $2\text{ mm} \times 2\text{ mm} \times 2\text{ mm}$); (4) regression of nuisance signals; and (5) denoise. Finally, the Brainnetome Atlas was used to parcellate fMRI volumes and compute regionally averaged fMRI signals. For each participant, we obtained an $N \times T$ matrix $S = [s_t]_{t=1,2,\dots,T}$ ($N = 246$ regions of Brainnetome Atlas, T is the time sampling number). We excluded those subjects with large head motion in any direction corresponding to $>3\text{ mm}$ or any rotation >3 . The head motion was also included as the concomitant variable.

All the DTI scans were pre-processed using the DiffusionKit Toolkit (<http://diffusion.brainnetome.org>).²³ DTI pre-processing comprised skull dissection, eddy current correction, registration, and normalization. Next the diffusion tensor indices' fractional anisotropy (FA) and mean diffusivity (MD) were calculated.¹⁹ After that, a brain network based on the number of fiber connections between brain regions was constructed using deterministic fiber tracking, and the brain structure network connection matrix (fiber number network, FN, sized $N \times N$) was obtained. Structural connectivity (SC) refers to the fiber connections between brain regions in the Brainnetome Atlas (Table S4). More detailed information is in [Method S2](#) and [Method S3](#).

2.3 | Function–structure coupling

To estimate the coupling of the SC and functional activities, we computed the regional-level SDI using the graph signal processing pipeline (https://www.github.com/gpreti/GSP_StructuralDecouplingIndex).¹¹ In brief, structural connectome harmonics were obtained by the eigen decomposition $LU = U\Lambda$ of the normalized Laplacian of the individual's structural connectome (FA, MD, and FN structural matrixes). This produces the u_k Laplace eigenvector, the so-called harmonic component, where each eigenvector is associated with an eigenvalue λ_k that can be interpreted as the graph frequency value. Through construction, u_k with low λ_k encodes low pattern frequencies that are more easily expressed on structural connectome, which represent global brain patterns along major geometric axes. Conversely, u_k with high λ_k encodes high pattern frequencies, capturing more complex and localized patterns. Then we projected the individual functional data of each time point onto individual structural harmonic and applied graphic signal filtering to decompose the active signal into two parts: one is expressed on the low-frequency structural harmonics (so it is more consistent with the structure) and the other is expressed on complementary high-frequency harmonics (therefore, more detached

RESEARCH IN CONTEXT

- Systematic review:** Authors reviewed the existing literature via PubMed searches, and found much research exploring the regions of structural and functional changes in Alzheimer's disease (AD). However, there is a lack of large-sample, multi-center studies investigating the relationship between structure and function in East Asian AD populations.
- Interpretation:** We found alterations in the structural decoupling index (SDI) of AD populations in the default mode network, frontal lobe, and subcortical regions, indicating changes in the structural–functional relationship due to metabolic abnormalities. SDI shows promising performance in identifying AD.
- Future directions:** These results suggest that SDI could serve as a biomarker for AD. Further analysis is warranted to explore the causal relationship between structural and functional changes in AD and to investigate how metabolic abnormalities lead to the dissociation of structure and function.

from the structure). The cutoff frequency C of the ideal low/high pass filter is defined by the equal energy division of the energy spectral density of each subject. The N square matrix U (low) contains the first C eigenmodules (columns of U) and complements $N-C$ zero columns. Conversely, the matrix U (high) includes the first C zero columns, followed by $N-C$ last eigenmodules. Therefore, the filtered signal is obtained from the formula (1) and the formula (2):

$$s_t^C = U_{(low)} U^T s_t \quad (1)$$

$$s_t^D = U_{(high)} U^T s_t \quad (2)$$

$$SDI = \frac{\sqrt{(s_{t1}^D)^2 + (s_{t2}^D)^2 + \dots + (s_{tn}^D)^2}}{\sqrt{(s_{t1}^C)^2 + (s_{t2}^C)^2 + \dots + (s_{tn}^C)^2}} \quad (3)$$

SDI is the ratio between the L2 norms of s_t^D and s_t^C across time and quantifies the absence of function–structure dependency in each region. Using this pipeline, we calculated regional-level individual SDIs for each metric (FA, MD, and FN; named FA-SDI, MD-SDI, and FN-SDI). Brain regions with an $SDI > 1$ suggest that their activity signals diverge more from the underlying structural pathways, resulting in lower coupling between SC and functional activities. Conversely, regions with an $SDI < 1$ exhibit the opposite relationship, where there is a stronger coupling between SC and functional activities.²⁴ (More details can be found in [Supplementary Materials](#).)

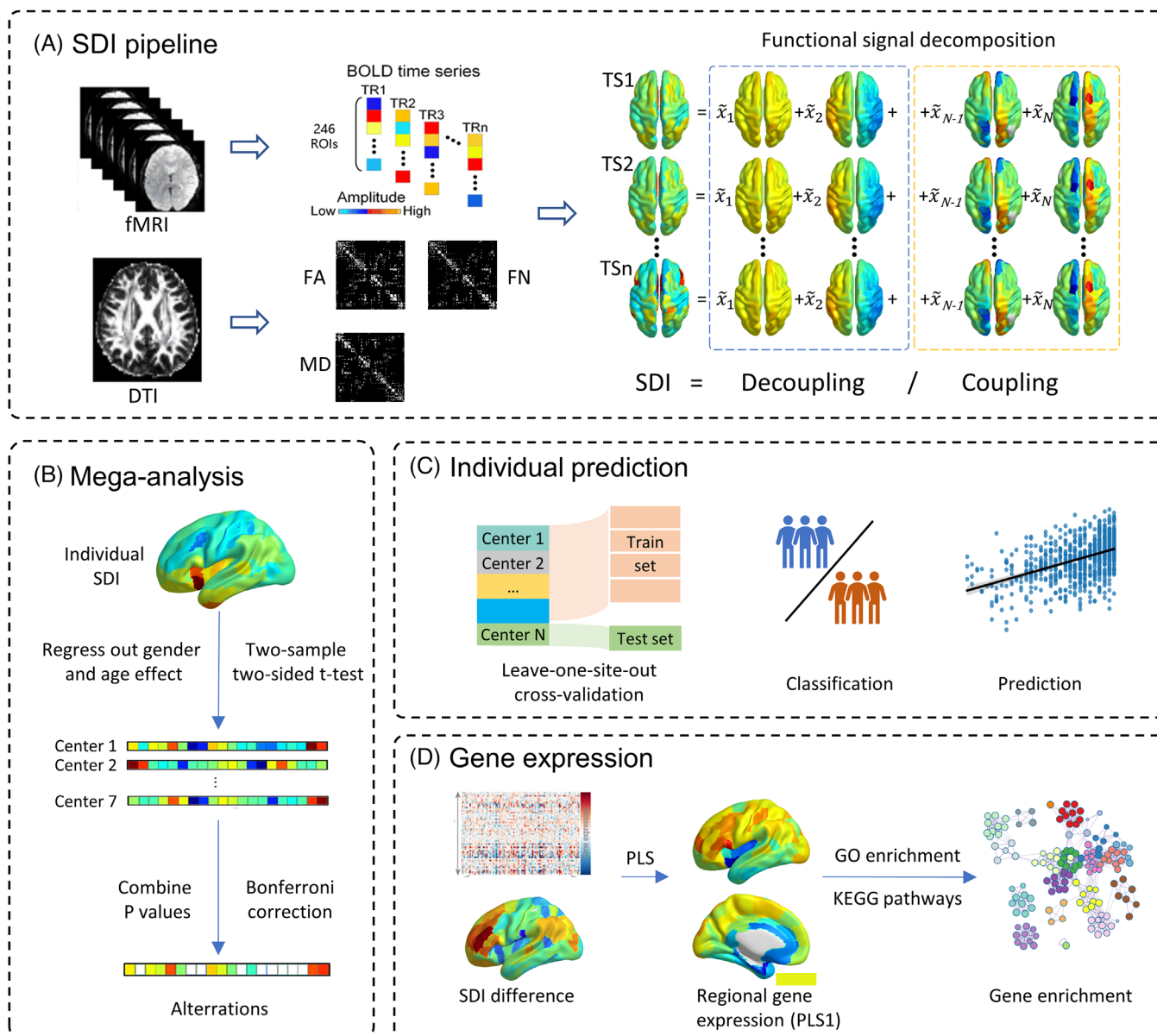


FIGURE 1 Method pipeline. (A) The pipeline for computing SDI. The rs-fMRI time series and structural matrix (FA, MD, and FN) were calculated based on the Brainnetome Atlas, and functional signals were then filtered into two components (coupling and decoupling from the structure) by applying ideal filters in the graph spectral domain. The SDI was the ratio between the norm of decoupling and coupling signal portions across time (see ¹¹). (B) Mega-analysis. A two-sample, two-sided *t*-test was performed to obtain the *p*-value for SDI metrics in each center after controlling for age and gender effects. The mega-analysis was applied to integrate results from seven centers, and the significantly altered regions were identified after multiple comparison corrections. (C) Individual prediction. The classification and prediction tasks are verified by leave-one-site-out cross-validation. (D) Gene expression. The partial least squares (PLS) regression was used to identify the weighted linear components of expression patterns for all 15,633 genes correlated with SDI alterations in connectome dynamics. FA, fractional anisotropy; FN, fiber number; MD, mean diffusivity; rs-fMRI, resting-state functional magnetic resonance imaging; SDI, structural decoupling index.

2.4 | Difference analysis for SDI between AD and NC

To reduce the site effects, we first applied two-sample, two-tailed *t*-tests for two groups at each site with the same scanning parameters. Then we performed a mega-analysis method to integrate the

multi-site results for each SDI metric (FA-SDI, MD-SDI, and FN-SDI). Therefore, the statistical results can be obtained more effectively by reducing the site effect via the mega-analysis framework compared to direct contrasts among the entire data set. Here, age and gender were controlled using the linear regression model. As suggested in previous studies, the Liptak-Stouffer *z*-score transform was used to

combine p -values across the seven sites, which has optimal power for combining probabilities in mega-analysis.^{20,22} Specifically, the p -values for each data set were transformed into z -scores using the inverse standard normal distribution. In particular, $z_i = \varphi^{-1}(1 - p_i/2)$, where φ^{-1} is the standard normal cumulative distribution function. The combined z -score was then computed using the Liptak–Stouffer formula²⁵:

$$z = \frac{\sum_{i=1}^k w_i z_i}{\sqrt{\sum_{i=1}^k w_i^2}} \quad (4)$$

where w_i is the square root of the sample size of data set i , and k is the number of data sets. Under the null hypothesis, the z -scores follow the standard normal distribution. Therefore, by converting the z -scores to p -values, we identified significant regions (FA-SDI, MD-SDI, and FN-SDI) that differed between the two groups. The Bonferroni correction was used to correct for multiple comparisons across the set of all 246 regions ($p < 0.05/246$).

2.5 | Associations between clinical scores, gene expression, and SDI alterations

To explore the biological basis of the SDI, we performed a Pearson's correlation between the SDI for each metric and cognitive ability, that is, the Mini-Mental State Examination (MMSE) scores and Montreal Cognitive Assessment (MoCA), in the AD and MCI groups with false discovery rate (FDR) correction ($p < 0.05$) after controlling the age, gender, and site effects.

A gene enrichment analysis was performed in the present study to explore the biological mechanism of the alteration of the SDI in AD. The gene expression data were obtained from the Allen Human Brain atlas (<http://human.brain-map.org/>) and were projected to the Brain-netome Atlas using the Abagen toolbox (<https://abagen.readthedocs.io/en/stable/>), resulting in a $236 \times 15,633$ gene expression matrix (10 of 246 regions did not identify related genes in the Abagen toolbox). We first performed a partial least squares (PLS) regression to investigate the association between the T-map of FA-SDI in AD versus NC and gene expression. Briefly, the statistical significance of the variance explained by the PLS components was tested using a permutation analysis ($n = 5000$) in which spatial autocorrelation was corrected. Then the PLS weight of each gene was transformed into a z -score value by dividing the weight by the standard deviation (SD) of the corresponding weights derived from 5000 bootstrapping instances. Finally, we effectively ranked 15,633 genes based on their corresponding weight value. After that, gene-set enrichment analysis was performed based on the top 1000 genes by the Metascape platform with FDR correction ($p < 0.05$) (<https://metascape.org/gp/index.html#/main/step1>).²⁶ Here, we provided only the association analysis based on the group difference map from the FA-SDI, as the results exhibited high similarity to those obtained from SDI analyses employing different metrics (Figure S1).

2.6 | Individual diagnostic status prediction

To evaluate whether the SDI can distinguish AD from NCs, we performed a binary classification by the most commonly used SVM model with features from 255 NCs and 284 ADs for each metric (FA-SDI, MD-SDI, and FN-SDI) with leave-one-site-out cross-validation framework, as in our previous studies.^{19,27} Briefly, one site was selected as the testing set for each time training and testing, and the others served as the training set. We used the 10-fold cross-validation in the training procedure to obtain the best parameters via the grid search framework. The performance of classification was evaluated using accuracy (ACC), sensitivity (SEN), specificity (SPE), and area under the receiver-operating characteristic (ROC) curve (AUC). To verify the clinical relativity of the classification model, we performed Pearson's correlation between individual risk score (evaluated by the classifier output) and MMSE score (in the ADs and MCIs) in the testing sets.

We further employed a support vector regression (SVR) model to predict the MMSE score for each participant (NC, MCI, and AD) via the same cross-validation framework as the classification analysis. The predicted MMSE performance based on the SDI was evaluated using the Pearson correlation coefficient between the actual and predicted MMSE scores.

3 | RESULTS

3.1 | Demographic and clinical characteristics

Among the NC, MCI, and AD groups there were no statistically significant differences in the ages and sex ratios. The MMSE score differed significantly among the NC, MCI, and AD groups, with $p < 0.001$ (Table S1).

3.2 | Alteration of the SDI in AD

The visual, sensory, motor, and auditory cortices showed low SDI values, whereas the frontal and temporal lobes presented higher SDI values. The SDI patterns showed a distinct spatial distribution, ranging from sensorimotor to higher-order functional regions (Figure 2A and Figure S2), which reflect a hierarchical structure of the brain.¹¹ We found that the global SDIs of NC after averaging 246 areas were significantly lower than those in AD in all three SDI metrics (Cohen's d , FA-SDI = -0.245 , MD-SDI = -0.246 , FN-SDI = -0.159) (Figure S3). In addition, the global SDIs were significantly lower in the MCI group than in the AD group in the FA-SDI and MD-SDI (Cohen's d , FA-SDI = -0.195 , MD-SDI = -0.194), (Table S5).

Compared with NCs, the patients with AD showed significant alteration in 27 areas (FA-SDI), 31 areas (MD-SDI), and 42 areas (FN-SDI) ($p < 0.05/N$, Bonferroni corrected for $N = 246$ comparisons), (Table S6 and Figure S4). As for FA-SDI, the AD group showed significantly higher SDI in the posterior superior temporal sulcus, insular gyrus,

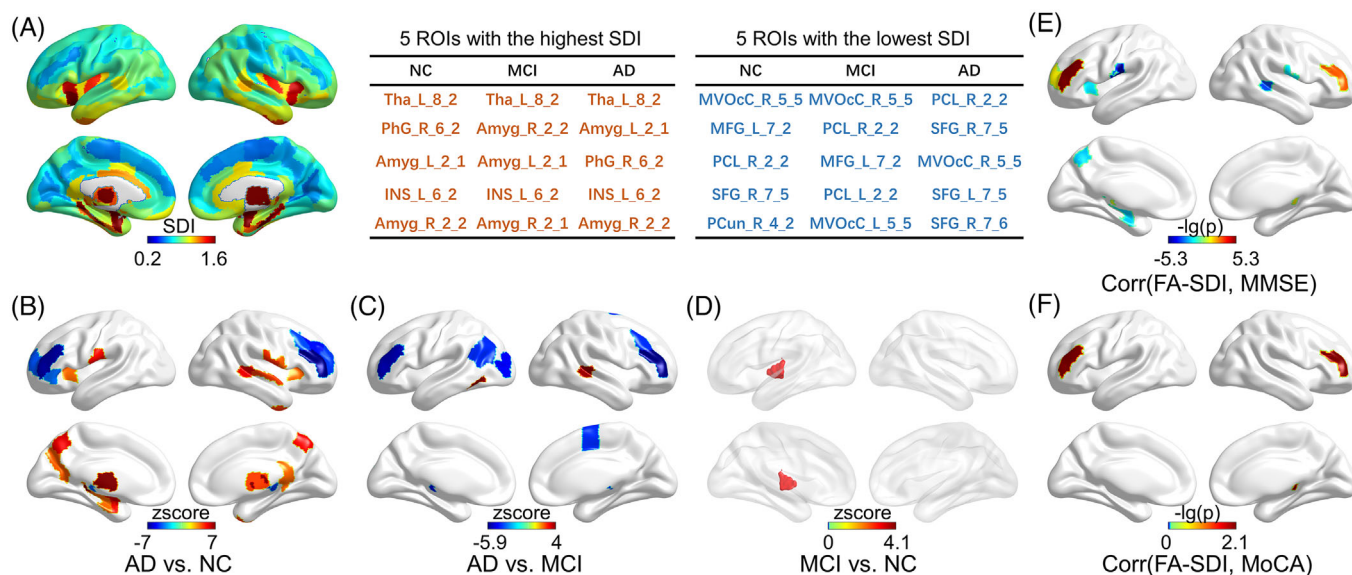


FIGURE 2 (A) Left: the FA-SDI pattern of NC; right tables showed the top five large (orange color) and low (blue color) FA-SDI scores with the associated NC, MCI, and AD groups. Statistically significant differences in FA-SDI between patients with AD versus NC (B) AD versus MCI, (C) and MCI versus NC. (D) The warmer and colder colors indicate higher and lower SDI measures in patients in the former group than in the latter group, respectively. The correlation map between altered FA-SDI and MMSE (E) and MoCA scores (F) in the AD and MCI patients with FDR correction ($p < 0.05$). AD, Alzheimer's disease; FA, fractional anisotropy; MCI, mild cognitive impairment; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; NC, normal control; SDI, structural decoupling index.

precuneus, hippocampus, amygdala, postcentral gyrus, and cingulate gyrus, whereas the group showed a significantly lower SDI in the middle frontal gyrus, and inferior frontal gyrus (Figure 2B). Compared with MCI in the FA-SDI, the AD group showed significantly higher SDI in the posterior superior temporal sulcus and inferior temporal gyrus, while showing significantly lower SDI in the superior frontal gyrus, middle frontal gyrus, inferior frontal gyrus, and thalamus (Figure 2C). We also found that compared to NCs, the MCI group only showed significantly higher FA-SDI in the thalamus (Figure 2D). The similarity-impaired patterns were found in MD-SDI and FN-SDI. (Details can be found in Figure S5.)

3.3 | Associations between SDI and clinical scores in the AD and MCI

Pearson's correlation analyses showed that the MMSE scores were significantly associated with 11 areas, 17 areas, and 12 areas in FA-SDI, MD-SDI, and FN-SDI, respectively. As for FA-SDI, significant positive correlations were found in the frontal lobe (superior frontal gyrus, middle frontal gyrus, inferior frontal gyrus), thalamus, and inferior parietal lobule, whereas significant negative correlations were found in the inferior temporal gyrus, postcentral gyrus, cingulate gyrus, hippocampus, posterior superior temporal sulcus, and insular gyrus (Figure 2E). The FA-SDI of the middle frontal gyrus also correlated significantly with the MoCA score (Figure 2F). Compared to FA-SDI, similar regions can also be found in MD-SDI, and part of the above areas were found in FN-SDI (Figure S6 and Table S6).

3.4 | Associations between SDI differences and gene expression

A significant spatial association was found between AD-related changes based on the SDI and nodal gene expression profiles (Figure 3A). The first PLS component (PLS1) accounted for a 20% variance in gene expression and significantly correlated with the T-map of SDI in AD versus NC ($r = 0.37$, $p = 3.6e-8$, corrected for Spin test), (Figure 3B and C). We found that the generation of precursor metabolites and energy (GO:0006091, $p_{FDR} = 9.26e-14$), inorganic ion transmembrane transport (GO:0098660, $p_{FDR} = 6.37e-12$), regulation of monoatomic ion transmembrane transport (GO:0034765, $p_{FDR} = 1.51e-10$), and mitochondrion organization (GO:0007005, $p_{FDR} = 2.47e-10$), (Figure 3D) are the most significantly correlated biological process for the alterations of the SDI. (Details can be found in Table S7).

3.5 | Multivariate classification and prediction based on SDI

The FA-SDI could be a robust neuroimaging biomarker to distinguish AD from NC with ACC = 0.77 (AUC = 0.864) via leave-one-site-out cross-validation (Figure 4A, Table 1, and Table S8). More significantly, we found a significant negative correlations ($r = -0.26$, $p = 6.8e-6$ for AD, $r = -0.18$, $p = 0.005$ for MCI, $r = -0.35$, $p = 1.9e-17$ for AD plus MCI) between the individual pseudo-probabilities and cognitive ability

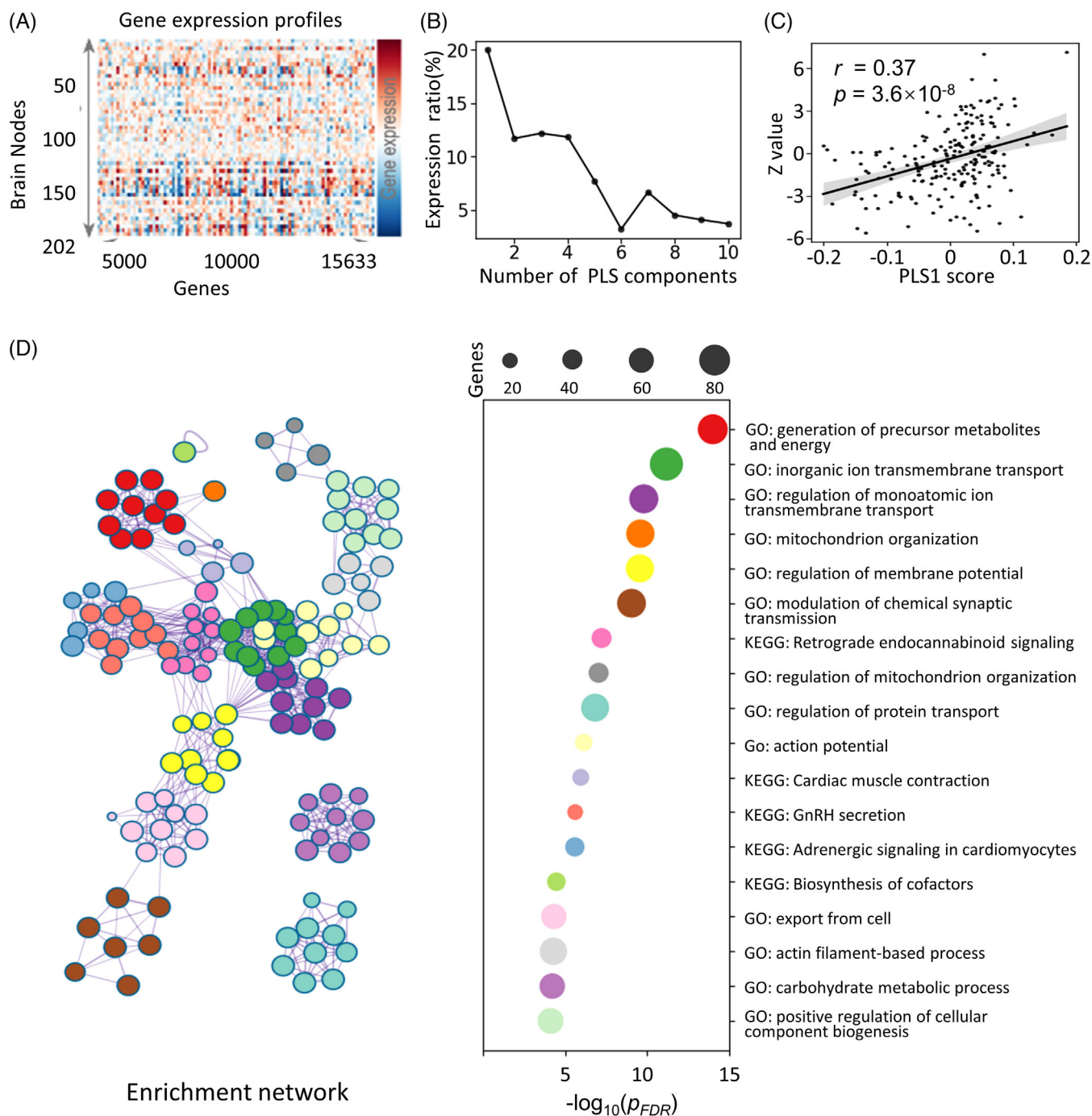


FIGURE 3 Association between AD-related alterations in module dynamics and gene expression profiles. (A) Gene expression profiles across brain nodes. Each row denotes the gene expression for each gene at a given brain node. (B) Explained ratios for the first 10 components obtained from the PLS regression analysis. Each component denotes a weighted linear combination of the expressions of all genes. (C) Spatial association between case-control differences in modular variability and PLS1 scores. Each dot represents a brain node. (D) Gene enrichment network for *PLS1* genes. The circle size represents the number of genes involved in the specific term. AD, Alzheimer's disease; PLS, partial least squares.

measured by MMSE (Figure 4B). Furthermore, the FA-SDI could also be a robust neuroimaging biomarker to predict the clinical cognitive score ($r = 0.45$, $p < 0.001$) via leave-one-site-out cross-validation (Figure 4C). Similar results can also be found in MD-SDI and FN-SDI compared to FA-SDI.

4 | DISCUSSION

Based on one of the largest multi-site multi-modal AD databases, the present study demonstrated that the SDI could serve as a neuroimaging measure for investigating the abnormal pattern of the brain's

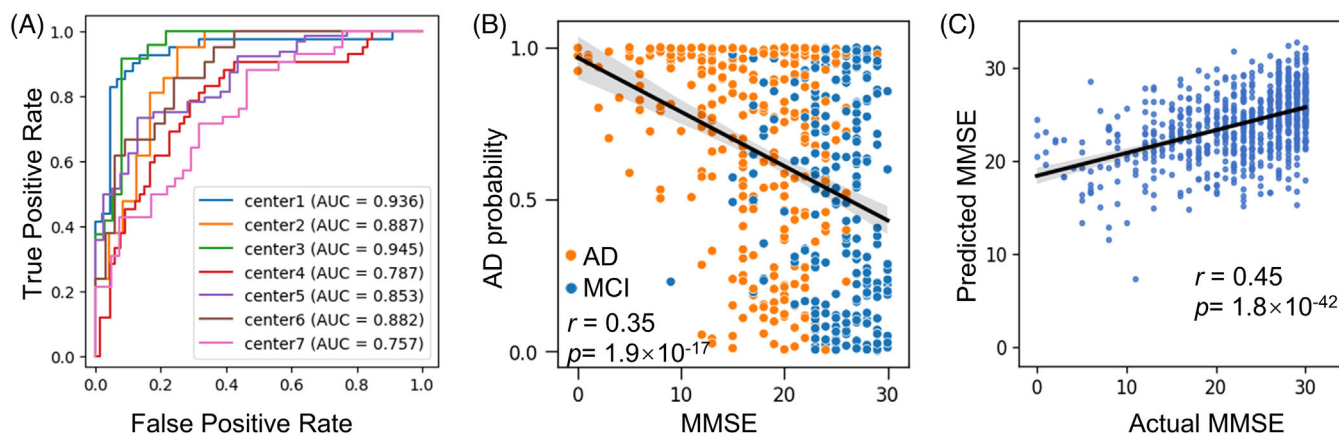


FIGURE 4 (A) The ROC and AUC of inter-site cross-validations. (B) Correlation between the AD probability of the test samples and MMSE scores. MMSE scores were z-scored within each data set and then pooled together. The results showed significant negative correlations ($r = -0.26$, $p = 6.8 \times 10^{-6}$ for AD, $r = -0.18$, $p = 0.005$ for MCI, $r = -0.35$, $p = 1.9 \times 10^{-17}$ for AD plus MCI) between the individual pseudoproabilities of AD and MCI subjects and the cognitive ability. (C) The correlation between predicted and actual MMSE scores of seven sites using leave-one-site-out cross-validation. AD, Alzheimer's disease; AUC, area under the ROC curve; MCI, mild cognitive impairment; MMSE, Mini-Mental State Examination; ROC, receiver-operating characteristic.

TABLE 1 Site-averaged performance of classification and regression.

	Classification AD from NC				Correlation between MMSE and AD risk score		Correlation between predicted MMSE and actual MMSE	
	AUC	ACC	SEN	SPE	<i>r</i>	<i>p</i>	<i>R</i>	<i>p</i>
FA-SDI	0.864	0.770	0.780	0.773	-0.346	1.91e-17	0.459	1.79e-42
MD-SDI	0.864	0.749	0.738	0.76	-0.323	1.63e-14	0.445	6.67e-40
FN-SDI	0.857	0.729	0.740	0.758	-0.127	3.27e-03	0.272	6.64e-15

Note: The site-averaged performances of individual prediction (AD vs NC classification and MMSE regression) using the SVM model and leave-one-site-out cross-validation with SDI features.

Abbreviations: ACC, accuracy; AD, Alzheimer's disease; AUC, area under the receiver-operating characteristic (ROC) curve; MMSE, Mini-Mental State Examination; SEN, sensitivity; SPE, specificity.

hierarchical structure in AD. The disease severity-associated regions largely overlapped with the altered brain areas, providing a potential biological explanation for cognitive decline in AD. Furthermore, the most common machine learning models with leave-one-site-out cross-validation accurately predicted individual diagnostic status. The classification results further support the clinical utility of SDIs as potential biomarkers for AD. These findings shed light on the relationship between structural-functional discrepancies, cognitive performance, and genetic pathways in AD.

In previous studies, the brain's hierarchical structure has been defined as the fundamental organizing principle for processing information, ranging from the primary cortex to the transmodal cortex.¹¹ Furthermore, it has been demonstrated that hierarchical organization elucidates how brain structure influences and constrains brain function, a concept known as modal decoupling, that is, transmodal cortex also occurs in a more separate pattern across multi-modal neuroimaging measures. The SDI has been successful in quantifying and characterizing the relationship between brain structure and function. Therefore, our study also demonstrated that the SDI could

be a valuable tool for understanding this relationship. The present study revealed SDI distribution ranging from sensory-motor regions to higher-order functional areas, indicating a pattern of structural constraint on function.²⁸ Of interest, it was consistent with brain hierarchical organization, that is, brain gradient, which ranged from strongly aligned "unimodal" sensory cortices to weakly aligned "transmodal" cortices in several previous studies.^{15,29-32} This hierarchical organization supports increasing levels of flexibility and dynamic processing from primary to high-order in the brain. In sensory-motor regions, the functional activity could be more directly supported by the underlying white matter pathways to react quickly to internal or external stimuli.³³ On the contrary, brain areas at the apex of the hierarchy may activate more in synchrony, not only as a consequence of direct signaling between them but also driven by inputs from the rest of the brain involved in polysynaptic indirect connections.³⁴ Consequently, the simultaneous functional activation of regions that are not structurally linked leads to a higher SDI. Therefore, the SDI provides a promising neuroimaging index to capture the brain's hierarchical organization.¹¹

Convergence evidence suggests that the brain is characterized by a hierarchical organization with functional–structural coupling.^{10,35,36} However, the divergence from coupling within one region and its pathological implications are poorly understood.³⁷ The present study revealed a similar coupling throughout the brain for both AD and NC groups, which confirmed that the dominant coupling pattern reflects a robust organizational principle of the brain. Meanwhile, we found a considerable number of regions with lower coupling in AD in the frontal lobe (superior frontal gyrus, middle frontal gyrus, inferior frontal gyrus) across three SDI metrics; these regions play a crucial role in executive function, attention, language, memory, and emotional regulation.³⁸ In addition, our findings showed an abnormally high coupling of patients with AD in the posterior superior temporal sulcus, insular gyrus, precuneus, hippocampus, and amygdala. The hippocampus, in particular, is known for its role in memory processing, and atrophy of the hippocampus is a prominent manifestation of AD.³⁹ These findings provide valuable insights into the regional variations in functional–structural coupling in AD, suggesting that alterations in coupling within specific brain regions may underlie the cognitive decline and memory impairments observed in AD. Understanding these regional differences in coupling can contribute to a deeper understanding of the neurobiological mechanisms of AD and potentially offer new targets for intervention and treatment.

The metabolic abnormalities and changes in mitochondrial function play essential roles in the development and progression of AD.^{40–43} Mitochondria are crucial organelles responsible for generating energy within cells, including neurons. Any disruptions in mitochondrial function can profoundly affect the energy supply to neurons and the transport of materials across the cell membrane. Abnormal energy metabolism can, in turn, impact the breakdown of the amyloid beta (A β) protein, an essential protein implicated in the formation of brain changes observed in AD.^{44,45} The accumulation of A β is associated with the development of AD pathology.^{46,47} Herein, abnormal energy metabolism can affect neuronal activity in the brain, leading to the loss of cognitive abilities, such as memory, language, and spatial orientation, in patients with AD.⁴⁰ These mental changes are then reflected in the SDI alterations, which reflect the dependence on brain structure and are highly related to the high-level cognitive activities of the brain. Abnormalities in this transport process can have wide-ranging effects on neurons, including signal transmission, energy metabolism, cell growth, and differentiation. These disruptions may contribute to programmed cell death and the overall neurodegeneration process seen in AD.⁴⁸ Understanding these underlying mechanisms is critical for developing therapeutic strategies to slow AD progression.

Despite the advancements presented herein, there are certain limitations. First, we all have confidence in the clinical diagnosis of patients diagnosed by experienced doctors from several famous hospitals in China. We should admit that the AD patients were clinically diagnosed, lacking biomarkers from cerebrospinal fluid (CSF) or positron emission tomography (PET), which introduces the possibility of misdiagnosis. In addition, longitudinal data sets for the MCI and AD groups should be included in future studies. Moreover, the coupling framework should

involve the structure covariance networks, that is, regional radiomics similarity network.^{49,50}

The present study characterized AD dementia-associated changes in the brain hierarchical structure by combining multisite rs-fMRI and DTI data. The disease-severity-associated pattern of the structural and functional connectivity coupling showed potential related to energy metabolism and mitochondrial organization pathways, providing valuable insights into the neurobiological changes in AD. These findings contribute to the broader understanding of the disease's pathophysiology in AD.

ACKNOWLEDGMENTS

This work was partially supported by the Science and Technology Innovation 2030 Major Projects (No. 2022ZD0211600), the Beijing Municipal Natural Science Foundation (No. 7244519), the Fundamental Research Funds for the Central Universities (No. 2021XD-A03), the National Natural Science Foundation of China (Nos. 62333002 and 82172018), the Beijing Nova Program (20220484177), the Science and Technology Project of Tianjin Municipal Health Committee (Grant No. TJWJ2022MS032), and Tianjin Key Medical Discipline (Specialty) Construction Project (Grant No. TJYXZDXK-052B).

CONFLICT OF INTEREST STATEMENT

The authors report no biomedical financial interests or potential conflicts of interest. Author disclosures are available in the [supporting information](#).

DATA AVAILABILITY STATEMENT

In the current study, all coding has been made publicly available through the GitHub repository maintained by YongLiuLab (<https://github.com/YongLiuLab>). These data sets had conditional acquisition from the corresponding author.

CONSENT STATEMENT

This study was consistent with the Principles of the Declaration of Helsinki and approved by the medical research ethics committee and institutional review board. Written informed consent was obtained from each enrolled subject or his/her authorized guardian.

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SUPPORTING INFORMATION

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How to cite this article: Sun Y, Wang P, Zhao K, et al. Structure–function coupling reveals the brain hierarchical structure dysfunction in Alzheimer's disease: A multicenter study. *Alzheimer's Dement*. 2024;20:6305–6315. <https://doi.org/10.1002/alz.14123>